

DSC 640 – Term Project Milestone 4

Nana Noda-Morgan

Summary of Analysis

Starting from the National Database of Childcare Prices, I calculated childcare burden as annual childcare cost divided by median household income. According to The Early Childhood Policy Team (2025), the national affordability benchmark is 7%, but what I found is that the national distribution is overwhelmingly to the right: nearly all counties exceed the line, with a dense cluster around 12-18%. State medians show the problem is especially acute in states like California and Washington, at roughly 16% and 18% respectively. Within those states, the major employer hubs including Los Angeles, Santa Clara, San Francisco, Sacramento, King, Spokane, sit at about 6-11 percentage points above the benchmark of 7%.

To prioritize action, I combined childcare burden with female unemployment to form a single priority matrix. Hubs like LA and Sacramento land in the “high burden and high unemployment” quadrant, which is a clear indicator of need and potential workforce benefit. Next, a “stipend needed to reach 7%” view translated the gap into dollars per child per month, giving executives a concrete measurement.

Finally, I modeled ROI with a clearly labeled synthetic employer-sites file, which included fields like headcount, salaries, turnover, uptake, etc. This allowed me to calculate savings from prevented turnover and reduced absenteeism, which help to offset program costs. I found that the net impact curve climbs steeply and peaks in the \$550-\$600/month range, with a break-even point of around \$350/month. This means a stipend at \$550-\$600 improves affordability for families *and* maximizes employer value.

Assumptions are surfaced and consistent across mediums: stipend paid monthly per child; uptake rate; retention lift (baseline + 15 percentage points); absenteeism improvement; and replacement cost at about 1.2x salary. These levers are adjustable in the Power BI ROI calculator and summarized in the executive brief so the audience can trace the case and effect.

Direction of Story

Family-related benefits are most often associated with better employee performance and satisfaction, leading to improved financial performance for the company

(Ranta & Ylinen, 2024). Armed with the data from my analysis, I aim to call for decision makers at these large corporations to approve an employer-sponsored childcare pilot in the form of a stipend. I will accomplish this by showing how childcare is systematically unaffordable in CA and WA hubs, and how a targeted stipend can bring families toward the 7% affordability benchmark while also delivering a compelling ROI.

Target Audience

My target audience are executives and decision makers, such as HR/Benefits leaders, at these large corporations based in California and Washington. A few specific corporations include Apple, NVIDIA, Starbucks, Microsoft, Google, Amazon, and Meta. These executives are very familiar with making decisions from data and comfortable with taking in information via presentations and simple dashboards. They expect to see concise visuals, transparent assumptions, and above all, a clear recommendation. My role is as a benefits consultant to these corporations, hired to design and manage competitive employee benefits programs.

Mediums Included & Why

- PowerPoint presentation
 - To show a compelling, linear story. Each slide will contain a new visualization that will persuade the audience to green light this pilot program. This is a very familiar medium for executives at these large corporations, and it allows for dialog along the way to clear up any questions, or dive into more details.
- One-page Executive Brief (PDF)
 - An executive brief will provide a digestible snapshot of the recommendation. This can accompany the presentation as a handout or emailed out. It should center around the core message of approving a stipend, with an ROI chart and assumptions clearly stated. This type of medium is simple to understand, even without a subject matter expert present.
- Interactive ROI Calculator/Dashboard
 - This will let decision makers test the assumptions and see the exact ROI for their use case. A dashboard is a perfect medium for decision makers to confirm numbers and be confident in their approval.

Design Decisions

I relied heavily on the Gestalt principles, as well as Cole Nussbaumer Knaflic's recommendations throughout all of the visualizations. I used left-aligned sentence titles that state the takeaway; I used neutral gray axes with light or no gridlines, I used a single accent blue color for emphasis, I used direct labeling instead of legends to reduce decoding, and I used consistent units and data formats. Benchmark and other lines are annotated where appropriate. Finally, the dashboard also mirrors this palette and removes any default chart components for a concise, executive-ready feel.

Ethical Considerations

Most important to mention is that the employer-sites dataset is synthetic and is used for scenario testing and communication. This synthetic dataset and the numbers derived from it could be mistaken for observed outcomes. I mitigate this with explicit 'synthetic' labeling, on-chart assumptions, and sensitivity views that show ranges rather than a specific number.

As for other data changes, the county burdens are derived directly from the National Database of Childcare Prices with light cleaning, which includes data type fixes and removing duplicates. Also, whenever I filter to CA and WA hubs or "Top x", that scope is stated in the slide or figure.

Finally, as far as legal guidelines are concerned, any real pilot needs to consider eligibility and nondiscrimination rules, as well as verifying tax implications, but those are post-approval steps that follow this presentation.

Lessons Learned

If I were to get a chance to do this project over again, I think I would spend more time upfront in creating a formatting file or function. It can contain things like centralized fonts, colors, axis, gridline rules, and annotation helpers so every chart I create can import the same style, so some of the finetuning steps aren't as repetitive.

What I enjoyed the most about this project was turning a messy, overwhelming dataset into a clean, decision-ready story. From state medians that set context, to the priority matrix that highlights the best pilot sites, and finally the net-impact curve that reveals a defensible stipend range. I was able to curate a story that made the call to action a no-brainer to pursue.

References

The Early Childhood Policy Team. (2025, July 16). Fact sheet: What to know about the Child Care for Working Families Act. Center for American Progress.

<https://www.americanprogress.org/article/fact-sheet-what-to-know-about-the-child-care-for-working-families-act/>

Ranta, M., & Ylinen, M. (2024). Employee benefits and company performance: Evidence from a high-dimensional machine learning model. *Management Accounting Research*, 64, 100876. <https://www.sciencedirect.com/science/article/pii/S104450052300046X>

OpenAI. (2025). ChatGPT (September 2025 version) [Large language model].

<https://chat.openai.com/>

I used ChatGPT to help me generate the synthetic employer dataset using the following prompt, and later manually updated some column values to fit my needs better:

“Please create a synthetic dataset in csv format with the following columns:

Employer, SiteName, County_Name, State_Abbreviation, County_FIPS_Code, Headcount, Percent_Parents_U6 (Parents with kids under 6), Avg_salary, Remote_Share, Baseline_Turnover_Rate, Cost_Per_Turnover, Retention_Lift_If_Stipend (expected reduction in turnover for parents), Uptake_Rate (parents who would use a stipend), Avg_children_per_parent

Include realistic records for major companies like Google, Apple, NVIDIA, Meta, Salesforce, Disney, Starbucks, Amazon, Microsoft, and Boeing, mapped to hub counties such as Santa Clara, San Mateo, San Francisco, Los Angeles, San Diego, Sacramento in CA; King, Snohomish, Spokane, Clark, Benton in WA. FIPs should be included so I can merge with another dataset.”